

GRADE 11 MATHEMATICS SAMPLE LESSON

CCSS: HSF-IF.C.7a; HSF-IF.B.4–6; HSF-BF.B.3; HSA-SSE.B.3

SMP: 2, 3, 4, 5, 7

LESSON LAUNCH

Objective: Students will identify, verify and solve the questions based on the patterns formed by quadratic equations.

Key Terms: Quadratic patterns, linear growth, second differences, nonlinear growths.

Context: Have you ever wondered that why everything in life never grows at a steady, predictable rate? If you earn or run at a fixed rate, your bank account or the distance covered will be in a straight line (that represents a linear function with a constant first difference). Now comes the cases where you drop a water balloon from a particular height, or the expanding area by the ripple produced in a pond. This types of growth accelerate with time.

Launch questions:

1. Imagine a cable hanging between two poles, the height of the cable from a particular point.
2. Imagine a ball being thrown in a particular direction, the distance covered by the ball.

Discussion prompt: What do you observe from all these changes that accelerate with time, can we find a predictable pattern?

Concept development: In this lesson, we will explore and understand the growth that accelerates with time by analyzing the structure of quadratic patterns, sequences and data tables by working with the second differences.

Mathematically, a sequence of terms $T_1, T_2, T_3, \dots$ is quadratic if it is defined by a second degree polynomial function, $T(n) = an^2 + bn + c$, $a \neq 0$, the second differences gives you a non-zero constant term $2a$.

Worked Examples:

Basic Understanding:

Identifying a quadratic sequence

Can you observe the pattern with the sequence 3, 8, 17, 30, 47...?

Is it linear, quadratic, or completely different?

Solution:

Step 1: let's calculate the first differences:

Terms	3	8	17	30	47
Difference	-	5	9	13	17

As these differences are changing, we can conclude that it's not linear

Step 2: now let's calculate the second differences

Terms	5	9	13	17
Difference	-	4	4	4

As we get the exactly same second difference, we can say that the sequence is quadratic.

Direct Application:

A table records the values generated by a quadratic relationship. Calculate the constant second difference and predict the value of y when $x = 5$

x	1	2	3	4	5
y	4	9	18	31	?

Solution:**Step 1:** the first diff = 5,9,13**Step 2:** the second diff = 4,4**Step 3:** the next first diff = $13+4 = 17$ **Step 4:** the next term = $31+17 = 48$ **Increased complexity:**A quadratic sequence has the missing term: 7, m , 11, 10, 7 ... Calculate the value of m .**Solution:****Step 1:** first diff = $(m - 7), (11 - m), -1, -3$ **Step 2:** second diff = $-3 - (-1) = -2$ **Step 3:** $(11 - m) - (m - 7) = -2$

$$18 - 2m = -2$$

$$2m = 20$$

$$m = 10$$

Application:

An outdoor theater is built with curved rows where the number of seats increases in each row from front to back :

Row 1: 6 seats , Row 2: 14 seats , Row 3: 26 seats , Row 4: 42 seats. Prove using second differences that the seating capacity per row forms a quadratic equation.

Solution:

Rows	1	2	3	4
Seats	6	14	26	42
First diff	-	8	12	16
Second diff	-	-	4	4

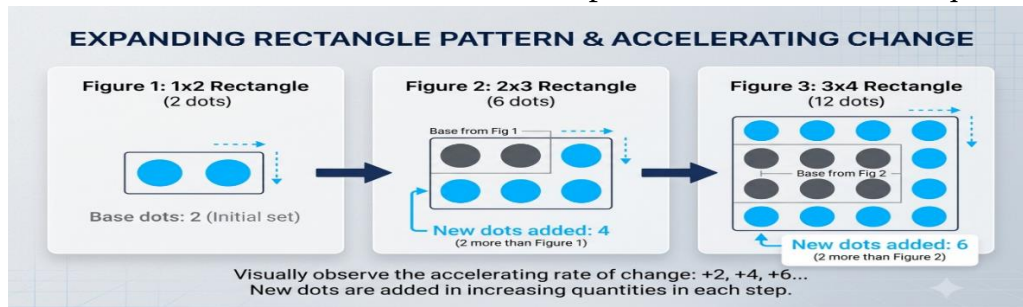
Before checking answers, you must remember that arithmetic mistakes in finding the first differences can ruin the second differences. To avoid this double-check the initial subtraction, especially when dealing with negative integers.

Guided Practice

1. Observe the sequence: 1, 4, 9, 16, 25 ... Is this sequence quadratic?
2. Consider the table and identify the type of the sequence:

x	1	2	3	4
y	5	12	23	38

3. Look at the illustration below with the dot pattern, what kind of sequence do we have?



4. A sequence has a constant second difference 4. Its last known term is 38, and the last known first difference is 15. Can you find the next first difference and the next term in the sequence?
5. Can we predict the pattern in the question 1 for any preferred number?

Independent practice

Foundation:

Find the first and second differences and classify the following as linear, quadratic or neither.

- 4,7,10,13,16 ...
- 2,5,10,17,26 ...
- 1,2,4,8,16 ...
- The sequence 0,3,8,15 ... is quadratic. What are the next two terms?
- Given any quadratic sequence with the first differences of 4,6,8 , what is the constant second difference?

On - level:

- A sequence is defined by the function $f(n) = n^2 + 3n$. Generate the first 4 terms.
- Prove that the sequence generated in the above question is quadratic.
- A quadratic pattern has a constant second difference of 6. If the first differences are 5,11,17 ... and the first term of the original sequence is 2, find the second and the third terms.
- James states that “the sequence 5,10,20,40 is quadratic because the differences keep getting larger at a fast rate. Is James mathematically correct? If not, then why?
- In a visual pattern of squares , the total number of square S at a particular step x is recorded as :

x	1	2	3	4
S	4	9	16	25

Write an explicit algebraic equation for $S(x)$.

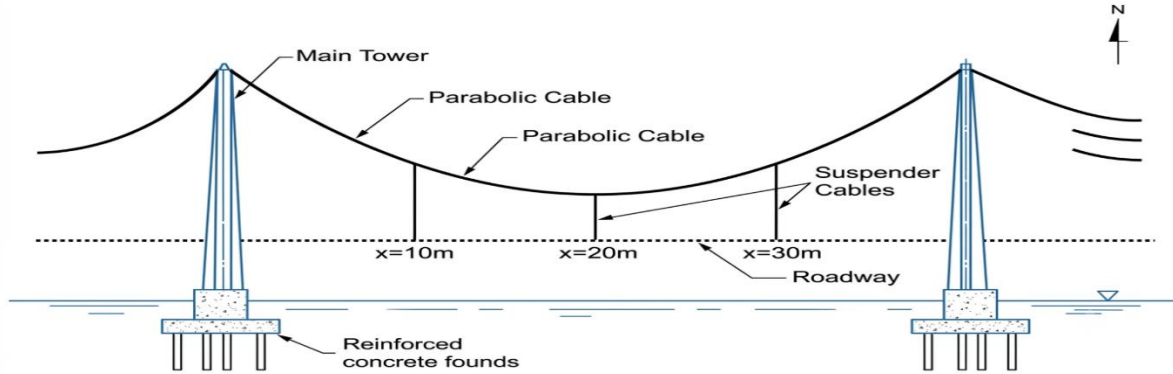
Brain builders

- A quadratic sequence is missing few terms: 3, ?, 19, 33, ?. Find the missing terms.
(Hint: let the missing terms be x and y and rely on the constant second differences)
- If a quadratic equation is given by $f(x) = ax^2 + bx + c$, prove algebraically that the second difference is always $2a$.
- Consider the sequence $f(x) = x^3$, calculate the first, second and the third differences and explain about the pattern that is being recorded.
- Create your own quadratic sequence of 5 terms where the constant second difference is -2 and the sequence eventually includes negative numbers.

5. A sequence is created by multiplying two consecutive integers i.e. $T(n) = n(n + 1)$. Prove that it is quadratic.

Application and mathematical reasoning

1. **Engineering:** Looking at the bridge diagram below, the height h (in meters) of the cable above the roadway is measured at 10 m intervals. The heights are 5m, 8m, 17m, 32m, 53m. Calculate the first and second difference to verify that whether the cable curve is quadratic or not?



2. **Physics:** Your classmate drops a heavy textbook from the roof of the gym. The total distance it falls (in meters) at the end of 1,2,3 and 4 seconds is recorded as 4.9m, 19.6m, 44.1m, 78.4m. Calculate the second differences. What real – world force does this constant value represent?

3. **Business:** A student – run customer car cleaning business tracks its daily profit as \$200, \$450, \$800, \$1250. Determine whether the growth is linear or quadratic.

4. **Economics:** if the car cleaning business stated above maintains any mathematical pattern, what will be the profits for days 6 and 7?

5. **Mathematical Argument:** A classmate argues that if a sequence has a constant first difference but the second difference is zero. He considers the sequence is quadratic. Would you agree or disagree? (*Construct a mathematical argument for a backup*)

LESSON CHECK

1. Which of the following is quadratic?

- | | |
|------------------|-------------------|
| a. 2,4,8,16 ... | b. 5,10,15,20 ... |
| c. 1,6,15,28 ... | d. 3,3,3,3... |

2. The sequence 4,11,22,37 ... is quadratic. What is the constant second difference?

- | | | | |
|------|-------|------|-------|
| a. 7 | b. 11 | c. 4 | d. 15 |
|------|-------|------|-------|

3. True or False: If the first differences of a sequence form a linear pattern, then the original sequence is quadratic.

4. A sequence has a second difference of 6. The last known first difference is 14, and the last known term of the sequence is 40. What will be the next term in the sequence?

5. The number of tiles used to build a series of expanding squares is modeled by $y = x^2$. Calculate the first and second differences for the first four terms.

ANSWER KEYS

GUIDED PRACTICE

1. Yes (second diff = 2)

2. Quadratic (second diff = 4)
3. Quadratic (second diff = 2)
4. Next diff = 19(15+4)
5. Yes ($f(x) = x^2$)

INDEPENDENT PRACTICE

Foundation:

1. Linear (first diff = 3: second diff = 0)
2. Quadratic (first diff = 3,5,7,9: second diff = 2,2,2)
3. Neither (first diff = 1,2,4,8: second diff = 1,2,4). This represents EXPONENTIAL GROWTH.
4. First diff = 3,5,7, the next diff = 9 , the next terms = 24(15+9) , 35(24+11)
5. 2

On – level:

1. 4,10,18,28,
2. First diff = 6,8,10 : second diff = 2,2
3. First term = 2 : second term = 7 (2+5) : third term = 18 (7+11)
4. James is mixing exponential growth with quadratic growth.
5. $S(x) = (x + 1)^2$

Brain builders:

1. Let the sequence be 3, x , 19,33, y

First diff = $(x - 3)$, $(19 - x)$, 14, $(y - 33)$

As the sequence is quadratic , the second difference is constant which gives

$$(19 - x) - (x - 3) = 14 - (19 - x)$$

$$22 - 2x = x - 5$$

$$3x = 27$$

$$x = 9$$

Therefore first diff = 6,10, 14

Now $y = 33 + 18 = 51$

Missing terms = 9,51

2. $f(x) = ax^2 + bx + c$

x	1	2	3
$f(x)$	$a + b + c$	$4a + 2b + c$	$9a + 3b + c$
First diff	-	$3a + b$	$5a + b$
Second diff	-	-	$2a$

3. $f(x) = x^3$

x	1	2	3	4	5
$f(x)$	1	8	27	64	125
First diff	-	7	19	37	61
Second diff	-	-	12	18	24
Third diff	-	-	-	6	6

4. Second diff = -2

First diff = -1,-3,-5,-7

Sequence = 10,9,6,1,-6

5. $T(n) = n(n + 1) = n^2 + n$

n	1	2	3	4
$T(n)$	2	6	12	20
First diff	-	4	6	8
Second diff	-	-	2	2

Applications and Mathematical reasoning

1. Yes

Height (m)	5	8	7	32	53
First diff	-	3	9	15	21
Second diff	-	-	6	6	6

2. Yes, second difference shows acceleration due to gravity

Distance (m)	4.9	19.6	44.1	78.4
First diff	-	14.7	24.5	34.3
Second diff	-	-	9.8	9.8

3. Yes

Profit (\$)	200	450	800	1250
First diff	-	250	350	450
Second diff	-	-	100	100

4. Day 5 = \$1800 (1250+550)

Day 6 = \$2450 (1800+650)

Day 7 = \$3200 (2450+750)

6. Agree. A linear equation $y = mx + b$ can be considered as a special case for $y = ax^2 + bx + c$ with $a = 0$. As the second difference is $2a$ and here as $a = 0$ we can have second difference as zero.

LESSON CHECK

1. c, first diff = 5,9,13 : second diff = 4,4

2. c, first diff = 7,11,15 : second diff = 4,4

3. True

4. Next first diff = 20 (14+6) : next term = 60 (40+20)

5. $y = x^2$

x	1	2	3	4
y	1	4	9	16
First diff	-	3	5	7
Second diff	-	-	2	2